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MRGeo: Robust Cross-View Geo-Localization of Corrupted Images via Spatial and Channel Feature Enhancement

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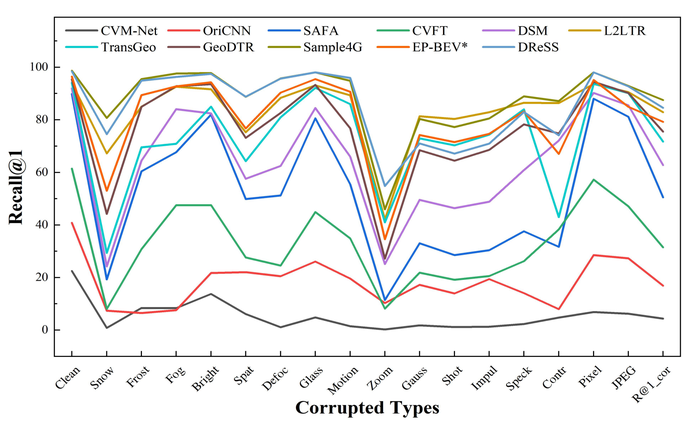
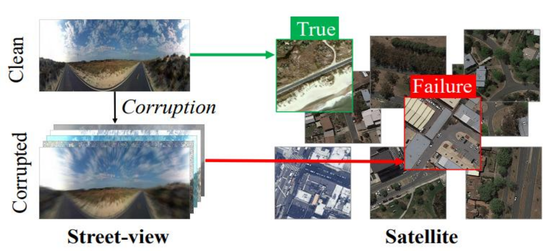
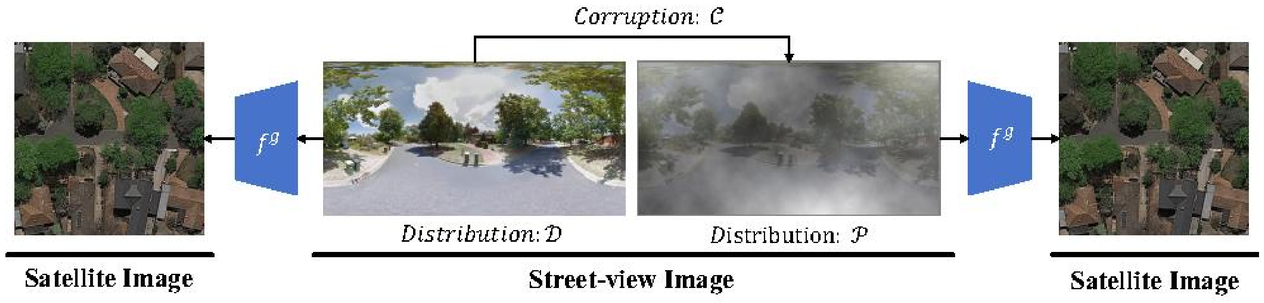
Abstract

While prior works have achieved nearly perfect performance on certain benchamrks, their robustness in real-world corrupted environments remains under-explored in **Cross-view geo-Localization (CVGL)** task. We employs a hierarchical defense strategy **MRGeo** that enhances the intrinsic quality of features and then enforces a robust geometric prior. **(1) Spatial Adaptive Representation Module (SARM)** that models global and local features in parallel and uses a dynamic gating mechanism to arbitrate their fusion based on feature reliability; **(2) Channel Calibration Module (CCM)** that performs compensatory adjustments by modeling multi-granularity channel dependencies to counteract information loss; **(3) Region-level Geometric Alignment Module (RGAM)** imposes a geometric structure on the final descriptors, ensuring coarse-grained consistency. **MRGeo** achieves superior performance improvement on robustness benchmarks and clean benchmarks.





Problem Definition



For Cross-view Geolocalization Localization on the ideal distribution $\mathcal{D}$:

$$\mathcal{D} \sim \{ d(f^g(I_i^g), d(f^g(I_i^s))\} < \{ d(f^g(I_i^g), d(f^g(I_j^s))\}, i \neq j;$$

For Cross-view Geolocalization Localization affected by various corruptions $\mathcal{C}$ (e.g., rain, snow, blur, noise) causing the distribution follow a more complex corrupted data distribution $\mathcal{P}$:

$$\mathcal{P} \sim \{ d(f^g(\mathcal{C}(I_i^g)), d(f^g(I_i^s))\} < \{ d(f^g(\mathcal{C}(I_i^g)), d(f^g(I_j^s))\}, i \neq j;$$

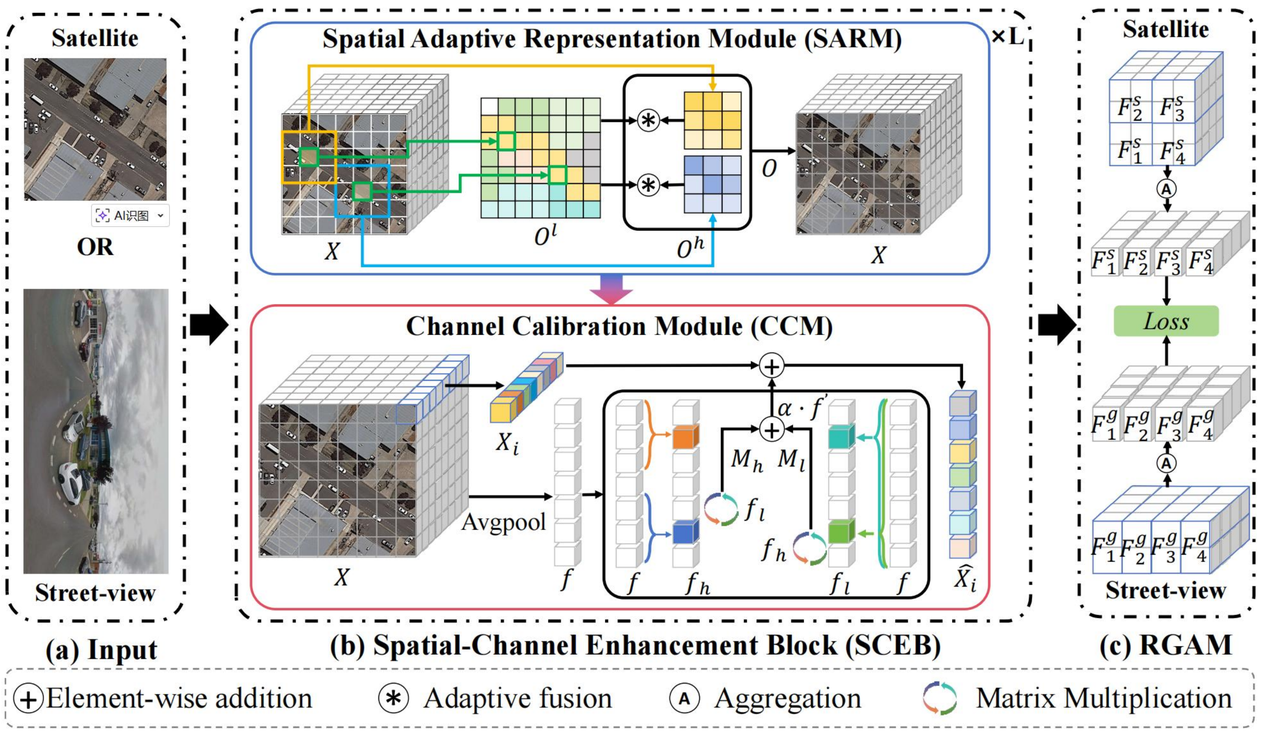
where I^g and I^s represent the street-view image and satellite image respectively.

Motivation

For the spatial domain: Image corrpution disrupts the fine local details that models rely on, while the relatively robust global structures are too coarse to distinguish similar scenes.

For the channel domain: Image corrpution introduces noise at the pixel level, which accumulates during network propagation and ultimately distorts the channel information of features, leading to semantic loss.

Method Overview



For Spatial Adaptive Representation Module, its most fundamental role is dynamically balancing global and local information. It intelligently assesses the reliability of local features, leveraging fine details when the image is clear and relying more on robust global structures when the image degrades:

$$O_i^l = (x_i W_q^l)\cdot(X W_k^l)^T\cdot(X W_v^l)^T$$
$$O_i^h = (x_i W_q^h)\cdot(\mathcal{N}_k(x_i) W_k^h)^T\cdot(\mathcal{N}_k(x_i) W_v^h)^T$$
$$O = O^l + \sigma(FC([O^l \cdot O^h])) \odot O^h$$

For Channel Calibration Module, its primary role is to compensate for the loss of channel information. By modeling multi-granularity channel dependencies, it calibrates and corrects distorted feature channels, thereby restoring crucial semantic information:

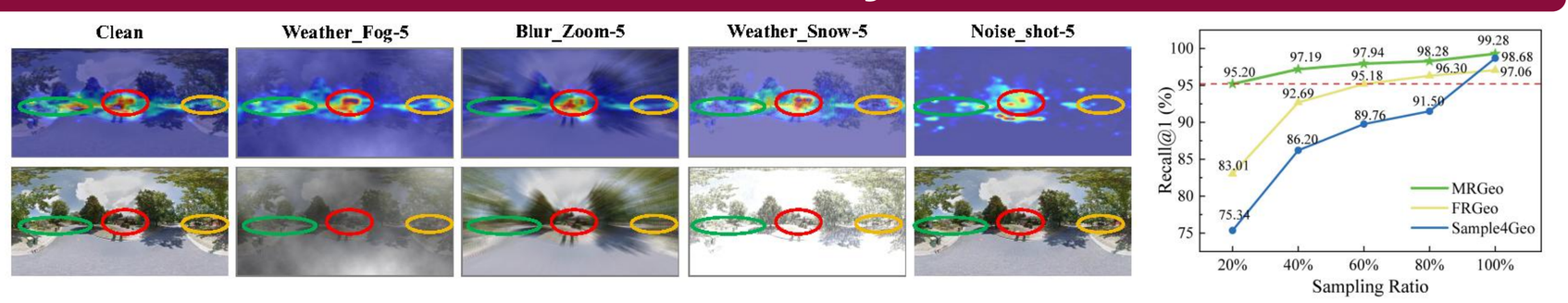
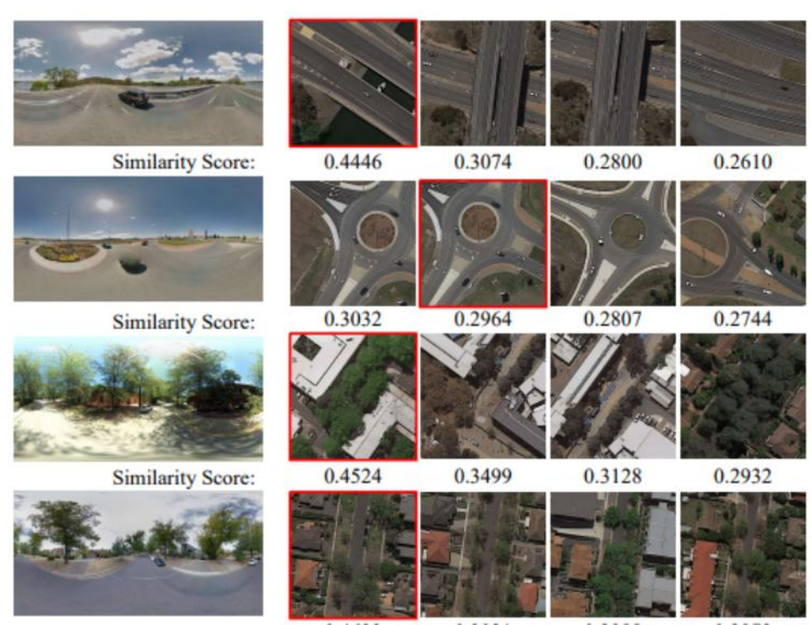
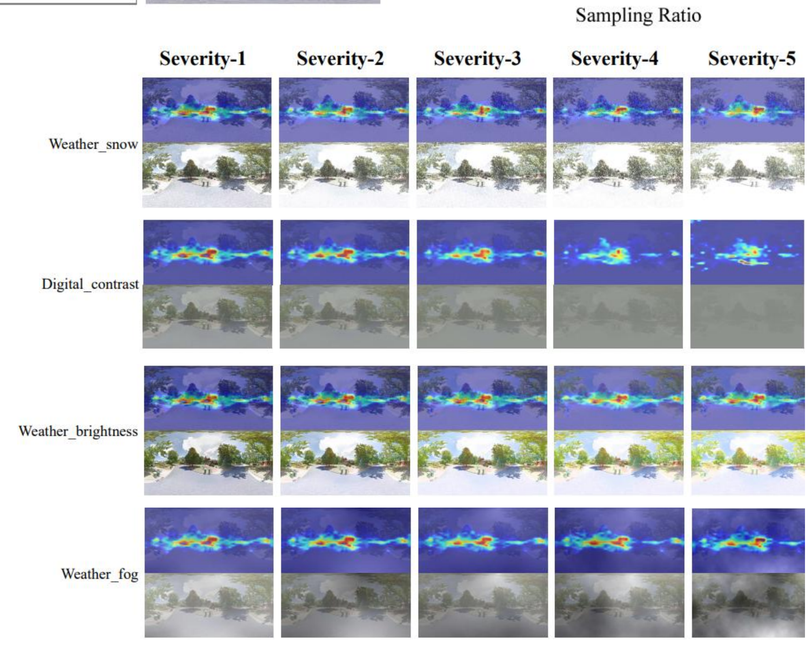
$$f_l = Conv1D(\text{Avgpool}(X))$$
$$f_h = Linear(\text{Avgpool}(X))$$
$$\hat{X}_i = \alpha(\underbrace{\mathcal{B}(f_h \cdot f_l^T)}_{M_h} + \underbrace{\mathcal{B}(f_l \cdot f_h^T)}_{M_l}) + X_i$$

For Region-level Geometric Alignment Module，it primarily enforces geometric consistency, thereby compelling the model to consider approximate geographic correspondence during matching and preventing severe spatial mismatches caused by feature ambiguity.:

$$f^i = [\mathcal{A}(F_1^i), \mathcal{A}(F_2^i), \mathcal{A}(F_3^i), \mathcal{A}(F_4^i)],\ where\ i \in \{g, s\}$$

Experiment Results & Visualization Analysis

Method	Venue	CVUSA-C-ALL				CVACT_val-C-ALL				CVACT_test-C-ALL			
		R@1	R@5	R@10	R@1%	R@1	R@5	R@10	R@1%	R@1	R@5	R@10	R@1%
CVM-Net	ICCV'18	6.09	16.05	23.14	52.51	–	–	–	–	–	–	–	–
OriCNN	CVPR'19	9.38	22.26	30.04	58.99	15.31	28.31	35.21	58.39	3.69	8.33	11.04	43.93
SAFA	NIPS'19	63.68	78.08	82.82	93.91	56.72	73.60	78.59	91.32	31.18	52.06	58.60	90.41
CVFT	AAAI'20	41.05	64.01	72.64	91.37	45.69	66.45	72.97	88.38	22.82	43.48	51.07	88.99
DSM	CVPR'21	75.27	86.26	89.42	95.07	70.04	82.81	85.86	93.51	47.13	68.41	73.52	93.18
L2LTR	NIPS'21	87.93	95.45	97.01	99.01	82.13	93.34	94.93	98.10	57.20	82.59	87.23	98.09
TransGeo	CVPR'22	82.72	91.95	94.03	97.92	74.04	86.19	89.10	94.98	52.18	74.35	78.99	95.03
GeoDTR	AAAI'23	84.64	93.29	95.01	98.24	77.40	88.95	91.28	95.91	52.87	78.84	83.17	95.84
Sample4G	ICCV'23	93.36	97.32	98.08	99.22	86.71	94.35	95.60	98.04	66.33	88.05	90.75	98.07
EP-BEV*	ECCV'24	84.33	94.38	96.06	98.39	82.07	92.19	94.09	97.52	59.25	83.12	87.02	97.49
MRGeo	AAAI'26	**95.09**	**97.99**	**98.48**	**99.38**	**90.45**	**96.50**	**97.33**	**98.91**	**71.16**	**92.24**	**94.21**	**98.80**

Method	Clean	CVACT_val-C															
		Weather					Blur				Noise				Digital			
		Snow	Frost	Fog	Bright	Spat	Defoc	Glass	Motion	Zoom	Gauss	Shot	Impul	Speck	Contr	Pixel	JPEG	$R@1_{cor}$
OriCNN	46.96	13.94	6.13	3.78	29.45	40.54	31.71	39.99	37.58	24.89	34.24	32.27	39.01	33.28	4.56	44.38	42.56	28.64
SAFA	81.03	20.03	31.66	33.19	66.99	45.60	39.83	72.87	49.86	4.62	48.66	43.68	48.82	51.61	15.91	76.90	75.83	45.38
CVFT	61.05	15.00	22.32	42.53	47.60	37.25	31.30	53.88	36.91	4.10	35.68	30.80	36.32	36.84	31.79	58.21	57.97	36.16
DSM	82.49	31.95	51.70	70.43	69.48	52.35	57.35	80.16	67.38	15.34	58.34	53.05	58.18	63.06	52.79	81.72	80.55	58.99
L2LTR	84.89	71.03	77.93	83.50	81.17	73.78	83.98	85.07	84.00	49.79	**82.20**	**81.19**	**82.98**	82.23	**79.15**	85.07	83.40	79.15
TransGeo	84.95	47.65	58.51	32.91	72.67	67.13	81.43	84.83	81.80	36.34	81.96	80.86	82.84	83.01	22.18	84.92	83.74	67.67
GeoDTR	86.21	48.24	71.74	83.26	84.60	61.39	79.11	85.51	73.44	8.26	75.44	73.99	77.06	80.23	55.48	86.01	85.19	70.56
Sample4G	90.81	78.69	84.68	86.97	88.89	84.97	89.17	90.22	89.03	50.00	67.33	62.10	70.02	73.10	51.99	90.26	89.52	77.93
EP-BEV*	88.91	69.43	78.67	81.55	86.08	79.77	87.52	88.55	87.36	38.15	60.61	53.66	63.56	63.44	49.09	88.77	86.92	72.70
MRGeo	**92.67**	**88.75**	**90.97**	**91.25**	**91.95**	**90.22**	**91.47**	**92.35**	**91.79**	**65.31**	79.29	73.97	82.47	**83.62**	72.71	**92.34**	**91.67**	**85.63**

#	SARM	CCM	RGAM	AugMix	CVACT_val-C-ALL				CVACT_test-C-ALL			
					R@1	R@5	R@10	R@1%	R@1	R@5	R@10	R@1%
1	×	×	×	×	86.77	94.55	95.88	98.31	65.84	88.32	91.03	98.30
2	×	×	✓	×	88.17	95.25	96.42	98.58	67.86	89.90	92.26	98.45
3	✓	✓	×	×	89.51	95.79	96.67	98.53	68.90	90.76	92.99	98.56
4	✓	✓	✓	×	**90.45**	**96.50**	**97.33**	**98.91**	**71.16**	**92.24**	**94.21**	**98.80**
5	2:1	✓	✓	×	89.77	95.90	96.84	98.46	70.40	91.37	93.38	98.50
6	1:1	✓	✓	×	89.27	95.85	96.80	98.65	69.16	90.82	93.86	98.58
7	1:2	✓	✓	×	81.87	91.95	93.71	97.34	59.55	83.51	86.91	97.24
8	✓	FC	✓	×	79.05	90.77	93.01	97.03	53.59	79.59	84.12	97.18
9	✓	XCiT	✓	×	89.18	95.78	96.78	98.64	69.87	90.89	93.09	98.62
10	✓	MFCM	✓	×	81.52	90.36	93.61	96.27	59.56	82.54	85.72	96.32
11	×	×	✓	✓	86.19	94.92	96.38	98.68	64.76	88.34	91.22	98.51